

Practice Exam 2

Problem 1

- (a)** Consider a single-phase to three-phase back-to-back converter configuration used in the United States with PFC and a grid peak current of $20\sqrt{2}$ A. What is the minimum DC-link voltage, and what is the corresponding phase voltage applied to the load at the maximum modulation index? The single-phase source voltage is $120 V_{\text{rms}}$. Assume a grid inductance of $\frac{1}{20\pi}$ H and a dc-link capacitance large enough to keep the dc-link voltage constant (no ripple).
- (b)** Repeat part (a) if the converter is used in an aircraft, where the grid frequency is 400 Hz, everything else remains the same. Tip: $|120 - j400| = 418$, and $418\sqrt{2} = 591$.

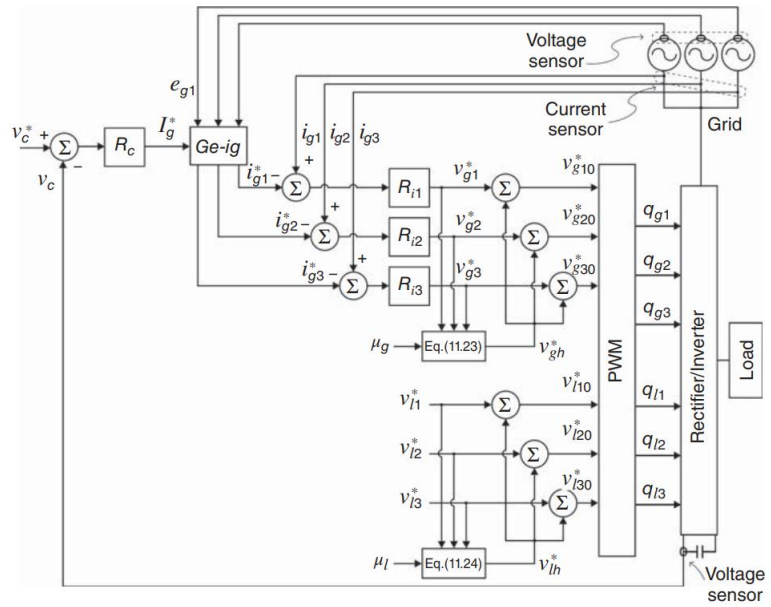
Problem 2

(a) A three-phase, four-wire ($3\Phi + N$) system fed by an inverter is typically employed when single-phase loads must be supplied from a three-phase system, such as in residential or mixed-use microgrids. It is also required in the presence of unbalanced loads, where a neutral conductor provides the necessary return path. This configuration is common in standalone or islanded systems, including solar-based microgrids, and is often adopted to ensure compatibility with standard distribution systems, such as 120/208 V networks in the United States. Propose a carrier-based PWM (CPWM) strategy by determining the reference pole voltages v_{10}^* , v_{20}^* , v_{30}^* , and v_{40}^* for a four-wire three-phase system with four legs. Assume that $v_h^* = -0.5(V_{max}^* + V_{min}^*)$ and derive expressions for V_{max}^* and V_{min}^* .

(b) The CPWM strategy is to be implemented digitally; therefore, expressions for the timers that define the gate signals for each switch must be determined.

Problem 3

The control block diagram of a three-phase back-to-back configuration is presented below. Since the regulators R_{i1} , R_{i2} , and R_{i3} are used to control AC variables, they cannot be implemented as simple PI controllers. Instead, they should be designed as resonant PI controllers. Modify the system as needed to allow the use of a standard PI controller $\left(k_p + \frac{k_i}{s}\right)$ while still ensuring zero steady-state error.



Problem 4

Consider a single-phase back-to-back configuration with a shared leg, meaning that one leg is common to both the input ($120 \text{ V}_{\text{rms}}$, 60 Hz) and the output, forming a three-leg topology. If the grid power factor is unity (with $\tilde{I}_g = 10 \text{ A}_{\text{rms}}$) and the load requires a frequency equal to that of the grid, what load power factor minimizes the converter's conduction losses? Assume a SPWM strategy operating in the linear region and $v_o^* = \cos(120\pi t - \pi/3)$.

Problem 5

Consider a three-phase to three-phase back-to-back converter supplying an electrical machine that initially consumes an active power of X . The machine subsequently reduces its active power consumption to $X/2$, while the measured grid current remains unchanged. Identify and explain the issue in the system. Under what conditions could this occur?